ROUND 4: BUSINESS PITCH

PrimeForge Manufacturing Ltd.

Data-Driven Quality & Defect Reduction Strategy

Executive Summary

Targeted analysis uncovers operational bottlenecks and actionable pathways to maximize manufacturing yield.

830

Line B Peak Defects

Highest vulnerability line

630

Line D Min Defects

Benchmark performance

3

Core Action Areas

Process, Line, Supplier

Key Analytical Findings

Root causes identified through exploratory data analysis and feature engineering.

Line Performance

Line B records maximum defects at 830, while Line D achieves optimal yield at 630 defects.

Process Deviations

Thermal and pressure spikes outside standard baselines directly correlate with higher rework frequencies.

Supplier Variations

Material quality fluctuates noticeably across suppliers, affecting final batch inspection outcomes.

Strategic Recommendations

Actionable measures to eliminate root causes and drive continuous improvement.

ACTION 01

Real-Time Sensor Monitoring

Deploy automated threshold alarms on Line B to prevent temperature spikes before defect formation.

ACTION 02

Supplier Quality Audits

Enforce rigorous incoming material inspections and periodic supplier performance audits.

ACTION 03

Predictive Maintenance

Track machinery operating hours to mitigate equipment wear and maintain calibration stability.

Questions & Discussion

Thank you for your time. Open for stakeholder review and feedback.